Project Report - Summer 2018

Rachana S Yajur

December 12, 2018

Scaling in Dynamical Quantum Phase Transitions

A project done at IISER Pune under the guidance of Prof. Sreejith GJ

Contents

1 Kramer-Wannier Duality in 2D Ising model **3**
1.1 High temperature expansion 3
1.2 Low temperature expansion 3
1.3 Resolving the differences between the two expansions 4

2 Transverse Field Ising Model **5**
2.1 Fermionic description 5
2.2 Bogoliubov transformation 6
2.3 Calculating the Loschmidt echo 7
2.4 Geometric approach 10
2.5 Rate function at the critical points: 13
2.6 Numerical simulations 16

3 Clock model $\mathscr{Z} - 3$ invariant spin chains **17**
3.1 Defining the Hamiltonian 17

1 Kramer-Wannier Duality in 2D Ising model

Consider the 2D ising model with periodic boundary conditions. Let the number of sites be $\mathscr{N}$. The total number of bonds (nearest neighbour interaction terms) will be $2\mathscr{N}$. The partition function can be calculated from the following expression[1] :

$$Z(K) = \sum_{s_i} \prod_{<ij>} e^{K s_i s_j} \qquad , K = \frac{1}{\beta T} \tag{1}$$

Where, $< ij >$ takes into account the nearest neighbour interactions and $\sum_{s_i}$ is the sum over all possible permutations over the values each s_i can take. Keeping in mind that $s_i s_j$ can only take the values ± 1, we get:

$$e^{K s_i s_j} = coshK + s_i s_j sinhK = coshK(1 + s_i s_j tanhK) \tag{2}$$

$$Z(K) = \sum_{s_i} \prod_{<ij>} (coshK + s_i s_j sinhK) = \sum_{s_i} \prod_{<ij>} coshK(1 + s_i s_j tanhK) \tag{3}$$

1.1 High temperature expansion

The product over $< ij >$ in (3) has $2\mathscr{N}$ number of terms resulting in :

$$Z(K) = \sum_{s_i} ((coshK)^{2\mathscr{N}} \prod_{<ij>} (1 + s_i s_j tanhK)) = 2^{\mathscr{N}} (coshK)^{2\mathscr{N}} \sum_{s_i} \prod_{<ij>} (1 + s_i s_j tanhK) \tag{4}$$

As there are $2^{\mathscr{N}}$ different combinations possible with $\mathscr{N}$ spins.

In the above expression, the term $\sum_{s_i} \prod_{<ij>} (1 + s_i s_j tanhK)$ can be evaluated in the following manner. Consider the product $s_i s_j$, considering all combinations of s_i and s_j, the product takes the values $+1$ twice and -1 twice. Hence when summed over all combinations it gives zero. Next, Equation 4 generates products of four spins $s_i s_j s_m s_n$. This, when summed over the various combinations results in zero. Same is the case with product over six spins. When considering product over 8 spins (coefficient of $tanh^4 K$ term), the sum over combinations does not vanish for cases where, the combination of spins form a closed loop, i.e., when each spin term is repeated even number of times, the sum over the combinations gives the number of possible closed loops with four bonds on the torus on which the 2D ising model with periodic boundary conditions is embedded. Similar procedure applies to the closed loops with 6,8,..,2n,.. bonds.

At high temperatures as $K \to 0$, higher order terms in $tanhK$ become less significant. Hence the high temperature expansion of the partition function can be approximated in the following way:

$$Z_{HT}(K) = 2^{\mathscr{N}} (coshK)^{2\mathscr{N}} (1 + \mathscr{N} tanh^4 K + 2\mathscr{N} tanh^6 K + 7\mathscr{N} tanh^8 K + \cdots) \tag{5}$$

1.2 Low temperature expansion

At low temperatures as $K \to \infty$, the partition function $Z(K) = \sum_{s_i} \prod_{<ij>} e^{K s_i s_j}$ can be expanded in the following manner:

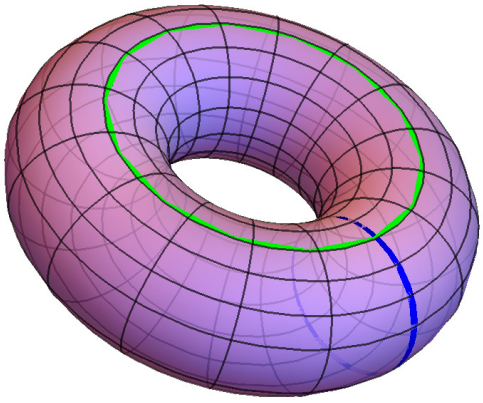


Figure 1:

The largest contribution to the partition function at low temperatures will be from the completely ordered state i.e., when all the spins are aligned either all up or all down. The next largest contribution will be when four bonds are broken (occurs when one spin is counter-aligned in the ordered state, as each spin is connected to four other spins, four bonds get 'broken' upon counter-aligning one spin). In counting the number of broken bonds possible for each spin configuration, we construct a dual lattice, where parallels can be drawn between set of broken bonds in real lattice and closed loops in the dual lattice. Hence the expansion of the partition function by counting the closed loops in the dual lattice is as given below.

$$Z_{LT}(K) = e^{2\mathscr{N}K}(1 + \mathscr{N}e^{-8K} + 2\mathscr{N}e^{-12K} + \cdot\cdot\cdot\cdot\cdot) \tag{6}$$

Notice that the coefficients of the expansion in e^{-K} is the same as that of the coefficients in the high temperature expansion as both involves counting of closed loops.

1.3 Resolving the differences between the two expansions

Even though the high temperature and low temperature expansions are empirically the same partition function, there appears to be a subtle difference in the two partition functions.

In the low temperature expansion, some closed loops considered in the dual lattice does not have physical correspondence to the broken bonds in the real lattice. These closed loops constitute the multiply connected (not-simply connected) loops on the torus. So this implies that $Z_{LT}(K)$ as in Equation 6 has extra terms counted into it.

Let $\mathscr{Z}_{LT}(K)$ be the actual partition function written in terms of low temperature expansion, which in principle must be equal to the high temperature expansion of the partition function $Z_{HT}(K)$.

$$Z_{HT}(K) = \mathscr{Z}_{LT}(K) = Z_{LT}(K) - C(K) \tag{7}$$

Where $C(K)$ is the unphysical counting over multiply connected loops in the dual lattice. Assuming square lattice with side length M where, $M^2 = \mathscr{N}$, the expression for $C(K)$ can be obtained in the following manner:

$$C(K) = e^{2\mathscr{N}K}(2e^{-2MK}(1 + (\mathscr{N} - M)e^{-8K}) + (2\mathscr{N} - 2M)e^{-12K} + (9\mathscr{N} - 9M) \tag{8}$$

$$\mathscr{Z}_{LT}(K) = e^{2\mathscr{N}K}(1+\mathscr{N}e^{-8K}+2\mathscr{N}e^{-12K}+..-2e^{-2MK}(1+(\mathscr{N}-M)e^{-8K})+(2\mathscr{N}-2M)e^{-12K}+(9\mathscr{N}-9M) \tag{9}$$

2 Transverse Field Ising Model

Transverse field ising Hamiltonian (periodic boundary condition):

$$H_{IM} = -J\sum_{n=1}^{N}\sigma_n^z\sigma_{n+1}^z - h\sum_{n=1}^{N}\sigma_n^x \tag{10}$$

where J is the coupling constant and h is the strength of the external magnetic field.

2.1 Fermionic description

By performing Jordan-Wigner transformations in the following way, we obtain the Hamiltonian in terms of fermionic operators.

Consider the fermionic operator, c_j at the jth site such that it satisfies the following commutation relations:

$$\{c_i, c_j^\dagger\} = \delta_{ij} \qquad [c_i, c_j] = 0 \tag{11}$$

Introducing the Majorana fermion operators with attached string operators such that the fermion antisymmetric properties under interchange between different sites is preserved is given by:

$$\psi_j^1 = \frac{1}{\sqrt{2}}[\prod_{l=1}^{n-1}\sigma_l^x]\sigma_n^y \tag{12}$$

$$\psi_j^2 = \frac{1}{\sqrt{2}}[\prod_{l=1}^{n-1}\sigma_l^x]\sigma_n^z \tag{13}$$

Where ψ_j^1 and ψ_j^1 also satisfy :

$$\psi_j^1 = \frac{c_j + c_j^\dagger}{2} \tag{14}$$

$$\psi_j^1 = \frac{c_j^\dagger - c_j}{2i} \tag{15}$$

From the above equations (3) and (4), we can express the pauli matrices in terms of Majorana fermion operators:

$$\sigma_j^x = -4i\psi_j^1\psi_j^2 \tag{16}$$
$$\sigma_j^z\sigma_{j+1}^z = 4i\psi_j^1\psi_{j+1}^2 \tag{17}$$

Now rewriting the Hamiltonian in terms of ψ^1 and ψ^2, we get :

$$H_{IM} = -\frac{J}{4i}\sum_{j=1}^{N}\psi_j^1\psi_{j+1}^2 - \frac{h}{4i}\sum_{j=1}^{N}\psi_j^1\psi_j^2 \tag{18}$$

Substituting Eqns (5) and (6) in (9) we obtain the Hamiltonian in terms of c_j's

$$H_{IM} = -J\sum_{j=1}^{N}(c_j^\dagger c_{j+1} + c_j^\dagger c_{j+1}^\dagger - c_j c_{j+1} - c_j c_{j+1}^\dagger) - 2h\sum_{j=1}^{N}c_j^\dagger c_j \tag{19}$$

Fourier transform of fermionic operators:

$$c_j = \frac{1}{\sqrt{N}}\sum_{k=-(N-1)/2}^{(N-1)/2} d_k e^{i\pi kj} \tag{20}$$

$$d_k = \frac{1}{\sqrt{N}}\sum_{j=-(N-1)/2}^{(N-1)/2} c_j e^{-i\pi kj} \tag{21}$$

New hamiltonian in terms of d_k's:

$$H_{IM} = -J\sum_{k\geq 0}^{\pi(N-1)}(d_k^\dagger d_k e^{ik} + d_k^\dagger d_{-k}^\dagger e^{ik} - d_k d_{-k}e^{-ik} - d_k d_k^\dagger e^{-ik}) - 2h\sum_{k\geq 0}^{2\pi(N-1)/2} d_k^\dagger d_k \tag{22}$$

Simplified form:

$$H_{IM} = -2\sum_{k\geq 0}^{\pi(N-1)/2}\begin{bmatrix}d_k^\dagger & d_{-k}\end{bmatrix}\begin{bmatrix}h+Jcos(k) & iJsin(k)\\ -iJsin(k) & -h-Jcos(k)\end{bmatrix}\begin{bmatrix}d_k\\ d_{-k}^\dagger\end{bmatrix} \tag{23}$$

2.2 Bogoliubov transformation

The eigen values of $M = \begin{bmatrix}h+Jcos(k) & iJsin(k)\\ -iJsin(k) & -h-Jcos(k)\end{bmatrix}$ are $\pm|B_k|$ where:

$$|B_k| = \sqrt{J^2+h^2+2Jhcos(k)} \tag{24}$$

Let $b_k = Jsin(k), c_k = h + Jcos(k)$

The eigen vectors of M are $\frac{1}{Norm}\begin{bmatrix}b_k\\ i(c_k+\sqrt{b_k^2+c_k^2})\end{bmatrix}$ and $\frac{1}{Norm}\begin{bmatrix}i(c_k+\sqrt{b_k^2+c_k^2})\\ b_k\end{bmatrix}$ where $\frac{1}{Norm}$ is the normalization factor with

$$Norm = \sqrt{2\sqrt{b_k^2+c_k^2}(c_k+\sqrt{b_k^2+c_k^2})} \tag{25}$$

Introducing a unitary matrix

$$U^\dagger = \frac{1}{Norm_k}\begin{bmatrix} b_k & i(c_k+\sqrt{b_k^2+c_k^2}) \\ i(c_k+\sqrt{b_k^2+c_k^2}) & b_k \end{bmatrix} \tag{26}$$

and rewriting the Hamiltonian:

$$H_{IM} = -\sum_{k\geq 0}^{\pi(N-1)/2} 2\begin{bmatrix} d_k^\dagger & d_{-k} \end{bmatrix} U^\dagger \begin{bmatrix} -|B_k| & 0 \\ 0 & |B_k| \end{bmatrix} U \begin{bmatrix} d_k \\ d_{-k}^\dagger \end{bmatrix} \tag{27}$$

Now let:
$\begin{bmatrix} d_k^\dagger & d_{-k} \end{bmatrix} U^\dagger := \begin{bmatrix} a_k^\dagger & a_{-k} \end{bmatrix}$ and $U\begin{bmatrix} d_k \\ d_{-k}^\dagger \end{bmatrix} := \begin{bmatrix} a_k \\ a_{-k}^\dagger \end{bmatrix}$ we get:

$$H_{IM} = -2\sum_{k\geq 0}^{\pi(N-1)/2} \begin{bmatrix} a_k^\dagger & a_{-k} \end{bmatrix} \begin{bmatrix} -|B_k| & 0 \\ 0 & |B_k| \end{bmatrix} \begin{bmatrix} a_k \\ a_{-k}^\dagger \end{bmatrix} \tag{28}$$

Simplifying the above equation we get:

$$H_{IM} = -2\sum_{k\geq 0}^{\pi(N-1)/2} (|B_k|a_k^\dagger a_k - |B_k|a_{-k}a_{-k}^\dagger) \tag{29}$$

Considering the fact that a_k's follow the same commutation relations as that of d_k's the hamiltonian can be further simplified into:

$$H_{IM} = -2\sum_{k=-\pi(N-1)/2}^{\pi(N-1)/2} |B_k|(a_k^\dagger a_k - \frac{1}{2}) \tag{30}$$

2.3 Calculating the Loschmidt echo

The BCS ground state is defined as

$$|\psi\rangle = \prod_{k>0} \frac{1}{Norm_k}(b_k + i(c_k+\sqrt{c_k^2+b_k^2})d_k^\dagger d_{-k}^\dagger)||0\rangle\rangle \tag{31}$$

Where $d_k^\dagger d_{-k}^\dagger$ is the Cooper pair creation operator. The reason for using this particularly is so that the momentum of the system is conserved. To simplify things, let $\frac{b_k}{Norm_k} = \frac{Jsin(k)}{\sqrt{2|B_k|(h+Jcos(k)+|B_k|)}} = cos\theta_k$ and $\frac{(c_k+\sqrt{c_k^2+b_k^2})}{Norm_k} = \frac{h+Jcos(k)+\sqrt{h^2+J^2+2hJcos(k)}}{\sqrt{2|B_k|(h+Jcos(k)+|B_k|)}} = sin\theta_k$ which reduces the Ground state to:

$$|\psi(0)\rangle = \prod_k (cos\theta_k + isin\theta_k d_k^\dagger d_{-k}^\dagger)||0\rangle\rangle \tag{32}$$

Time evolution of the ground state is carried out with the time evolution operator generated by the modified Hamiltonian.

$$|\psi(t)\rangle = e^{-iH'_{IM}t}\prod_k (cos\theta_k + isin\theta_k d_k^\dagger d_{-k}^\dagger)||0\rangle\rangle \tag{33}$$

Where $H_{IM}^{'}$ is given by:

$$H_{IM}^{'} = -2 \sum_{k\geq 0}^{\pi(N-1)/2} \begin{bmatrix} d_k^{\dagger} & d_{-k} \end{bmatrix} \begin{bmatrix} h' + J'cos(k) & iJ'sin(k) \\ -iJ'sin(k) & -h' - J'cos(k) \end{bmatrix} \begin{bmatrix} d_k \\ d_{-k}^{\dagger} \end{bmatrix} \tag{34}$$

In this case $U'^{\dagger}$ becomes from (9):

$$U'^{\dagger} = \begin{bmatrix} cos\theta_k^{'} & isin\theta_k^{'} \\ isin\theta_k^{'} & cos\theta_k^{'} \end{bmatrix} \tag{35}$$

$$H_{IM}^{'} = - \sum_{k\geq 0}^{\pi(N-1)/2} 2 \begin{bmatrix} d_k^{\dagger} & d_{-k} \end{bmatrix} U'^{\dagger} \begin{bmatrix} -|B_k'| & 0 \\ 0 & |B_k'| \end{bmatrix} U' \begin{bmatrix} d_k \\ d_{-k}^{\dagger} \end{bmatrix} \tag{36}$$

$$H_{IM}^{'} = - \sum_{k\geq 0}^{\pi(N-1)/2} 2 \begin{bmatrix} d_k^{\dagger} & d_{-k} \end{bmatrix} \begin{bmatrix} cos\theta_k^{'} & isin\theta_k^{'} \\ isin\theta_k^{'} & cos\theta_k^{'} \end{bmatrix} \begin{bmatrix} -|B_k'| & 0 \\ 0 & |B_k'| \end{bmatrix} \begin{bmatrix} cos\theta_k^{'} & -isin\theta_k^{'} \\ -isin\theta_k^{'} & cos\theta_k^{'} \end{bmatrix} \begin{bmatrix} d_k \\ d_{-k}^{\dagger} \end{bmatrix} \tag{37}$$

$$H_{IM}^{'} = - \sum_{k\geq 0}^{\pi(N-1)/2} 2 \begin{bmatrix} cos\theta_k^{'} d_k^{\dagger} + isin\theta_k^{'} d_{-k} & cos\theta_k^{'} d_{-k} + isin\theta_k^{'} d_k^{\dagger} \end{bmatrix} \begin{bmatrix} -|B_k'| & 0 \\ 0 & |B_k'| \end{bmatrix} \begin{bmatrix} cos\theta_k^{'} d_k - isin\theta_k^{'} d_{-k}^{\dagger} \\ cos\theta_k^{'} d_{-k}^{\dagger} - isin\theta_k^{'} d_k \end{bmatrix} \tag{38}$$

Loschmidt amplitude is given by:

$$\mathscr{G}(t) = \langle \psi(0) | \psi(t) \rangle \tag{39}$$

$$\mathscr{G}(t) = \langle \psi(0) | e^{-iH_{IM}^{'}t} | \psi(0) \rangle \tag{40}$$

$$\langle \psi(0) | e^{-iH_{IM}^{'}t} | \psi(0) \rangle = \langle \langle 0 || \prod_k (cos\theta_k - isin\theta_k d_{-k} d_k) e^{-iH_{IM}^{'}t} \prod_k (cos\theta_k + isin\theta_k d_k^{\dagger} d_{-k}^{\dagger}) || 0 \rangle \rangle \tag{41}$$

$$e^{-iH_{IM}^{'}t} = \prod_{k>0} (e^{it|B_k'| a_k^{'\dagger} a_k'} e^{it|B_{-k}'| a_{-k}^{'\dagger} a_{-k}'}) \tag{42}$$

A useful simplification:

$$e^{it|B_k'| a_k^{'\dagger} a_k'} = 1 + a_k^{'\dagger} a_k' (e^{i|B_k'|t} - 1) \tag{43}$$

$$e^{it|B_k'| a_k^{'\dagger} a_k'} e^{it|B_{-k}'| a_{-k}^{'\dagger} a_{-k}'} = (1 + a_k^{'\dagger} a_k' (e^{i|B_k'|t} - 1))(1 + a_{-k}^{'\dagger} a_{-k}' (e^{i|B_{-k}'|t} - 1)) \tag{44}$$

$$= 1 + a_k^{'\dagger} a_k' (e^{i|B_k'|t} - 1) + a_{-k}^{'\dagger} a_{-k}' (e^{i|B_k'|t} - 1) + a_k^{'\dagger} a_k' a_{-k}^{'\dagger} a_{-k}' (e^{i|B_k'|t} - 1)^2 \tag{45}$$

From (11) we know that:

$$a_k^{'\dagger} a_k' = (cos\theta_k' d_k^{\dagger} + isin\theta_k' d_{-k})(cos\theta_k' d_k - isin\theta_k' d_{-k}^{\dagger}) \tag{46}$$

In obtaining the Loschmidt amplitude, the following simplifications are utilized:

$$a'_k(cos\theta_k + isin\theta_k d^\dagger_k d^\dagger_{-k}) = (cos\theta'_k d_k - isin\theta'_k d^\dagger_{-k})(cos\theta_k + isin\theta_k d^\dagger_k d^\dagger_{-k})$$
$$= cos\theta'_k cos\theta_k d_k + isin(\theta_k - \theta'_k)d^\dagger_{-k}$$

$$\langle\langle 0||(cos\theta_k - isin\theta_k d_{-k} d_k)a^{'\dagger}_k a'_k(cos\theta_k + isin\theta_k d^\dagger_k d^\dagger_{-k})||0\rangle\rangle = sin^2(\theta_k - \theta'_k) \tag{47}$$

$$a'_{-k}(cos\theta_k + isin\theta_k d^\dagger_k d^\dagger_{-k}) = (cos\theta'_k d_{-k} + isin\theta'_k d^\dagger_k)(cos\theta_k + isin\theta_k d^\dagger_k d^\dagger_{-k})$$
$$= cos\theta'_k cos\theta_k d_{-k} - isin(\theta_k - \theta'_k)d^\dagger_k$$

$$\langle\langle 0||(cos\theta_k - isin\theta_k d_{-k} d_k)a^{'\dagger}_{-k} a'_{-k}(cos\theta_k + isin\theta_k d^\dagger_k d^\dagger_{-k})||0\rangle\rangle = sin^2(\theta_k - \theta'_k) \tag{48}$$

$$a'_k a^{'\dagger}_{-k} = (cos\theta'_k d_k - isin\theta'_k d^\dagger_{-k})(cos\theta'_k d^\dagger_{-k} - isin\theta'_k d_k)$$
$$= d_k d^\dagger_{-k}$$

$$\langle\langle 0||(cos\theta_k - isin\theta_k d_{-k} d_k)a^{'\dagger}_k a'_k a^{'\dagger}_{-k} a'_{-k}(cos\theta_k + isin\theta_k d^\dagger_k d^\dagger_{-k})||0\rangle\rangle = sin(\theta_k - \theta'_k)^2 \tag{49}$$

Inserting the results obtained in (30),(31) and (32) into (28), the following expression for the Loschmidt amplitude is obtained:

$$\mathscr{G}(t) = \prod_{k>0}(1 + 2sin^2(\theta_k - \theta'_k)(e^{it|B'_k|} - 1) + sin(\theta_k - \theta'_k)^2(e^{it|B'_k|} - 1)^2) \tag{50}$$
$$= \prod_{k>0}(1 + sin^2(\theta_k - \theta'_k)(e^{2it|B'_k|} - 1)) \tag{51}$$
$$= \prod_{k>0}(cos^2(\theta_k - \theta'_k) + sin^2(\theta_k - \theta'_k)e^{2it|B'_k|}) \tag{52}$$

The Loschmidt echo is defined by:

$$\mathscr{L}(t) = |\mathscr{G}(t)|^2 \tag{53}$$

The rate function is given by:

$$\lambda(t) = -lim_{N\to\infty}\frac{1}{N}log\mathscr{L}(t) \tag{54}$$

For finite N, the rate function takes the following form:

$$\lambda(t) = -\frac{2}{N}\sum_k log(|(cos^2(\theta_k - \theta'_k) + sin^2(\theta_k - \theta'_k)e^{2it|B'_k|})|) \tag{55}$$

Rate function in the thermodynamic limit:

$$\lambda(t) = -\int_0^\pi \frac{dk}{\pi} log(|(cos^2(\theta_k - \theta'_k) + sin^2(\theta_k - \theta'_k)e^{2it|B'_k|})|) \tag{56}$$

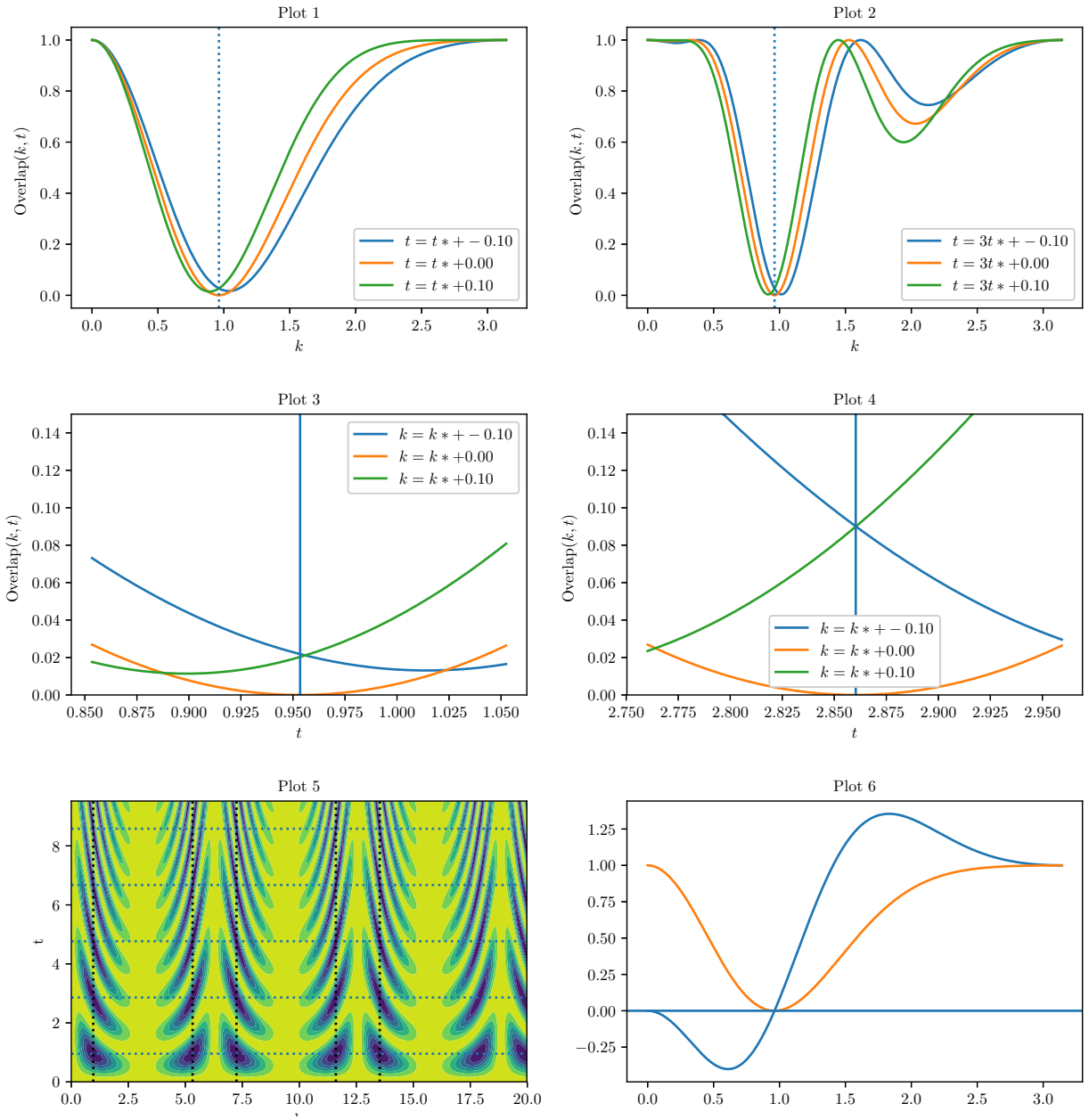


Figure 2: Investigating the Loschmidt amplitude

2.4 Geometric approach

Consider the vector $\vec{B_k} = (0, Jsin(k), h + Jcos(k))$ and $\vec{\sigma} = (\sigma^x, \sigma^y, \sigma^z)$. $\vec{B_k} \cdot \vec{\sigma} = M$ where M is the matrix of the Hamiltonian when expressed in terms of fermion operators in momentum space.

Let $\vec{B_k}$ represent a magnetic field vector. The ground state vector associated with this magnetic field is the eigen state of M with the eigenvalue $-\sqrt{h^2 + J^2 + 2hJcos(k)}$. The eigen state is

$$|\phi\rangle = \frac{1}{Norm_k} \begin{bmatrix} Jsin(k) \\ i(h + Jcos(k) + \sqrt{h^2 + J^2 + 2hJcos(k)}) \end{bmatrix} = \begin{bmatrix} cos(\theta) \\ isin(\theta) \end{bmatrix} \tag{57}$$

$$|\phi\rangle = \begin{bmatrix} cos(\theta) \\ isin(\theta) \end{bmatrix} = cos\frac{\alpha}{2}\begin{bmatrix}1\\0\end{bmatrix} + e^{i\pi/2}sin\frac{\alpha}{2}\begin{bmatrix}0\\1\end{bmatrix}$$

$$\text{where } \theta = \alpha/2$$

Let $\begin{bmatrix}1\\0\end{bmatrix} := |0\rangle$ and $\begin{bmatrix}0\\1\end{bmatrix} := |1\rangle$ which gives $|\phi\rangle = cos\frac{\alpha}{2}|0\rangle + e^{i\pi/2}sin\frac{\alpha}{2}|1\rangle$

Now consider the changed magnetic field, $\vec{B_k'} = (0, h' + J'cos(k), J'sin(k))$. The time evolution operator formed from this magnetic field brings about rotation to the initial ground state vector

$$\begin{aligned}
|\phi(t)\rangle &= e^{it\gamma\vec{B_k}'\cdot\vec{\sigma}}|\phi(0)\rangle \\
&= e^{it\gamma\vec{B_k}'\cdot\vec{\sigma}}(cos\frac{\alpha}{2}|0\rangle + e^{i\pi/2}sin\frac{\alpha}{2}|1\rangle) \\
\vec{B_k}'\cdot\vec{\sigma} &= J'sink\sigma^y + (h' + J'cosk)\sigma^z \\
\vec{B_k}'\cdot\vec{\sigma}\,|0\rangle &= -iJ'sink|1\rangle + (h' + J'cosk)|0\rangle \\
\vec{B_k}'\cdot\vec{\sigma}\,|1\rangle &= iJ'sink|0\rangle - (h' + J'cosk)|1\rangle \\
e^{it\gamma\vec{B_k}'\cdot\vec{\sigma}} &= 1cos(\gamma t|B_k'|) + i\frac{\vec{B_k'}\cdot\vec{\sigma}}{|B_k'|}sin(\gamma t|B_k'|)
\end{aligned}$$

$$\begin{aligned}
|\phi(t)\rangle = (cos\frac{\alpha}{2}cos(\gamma t|B_k'|) + i\frac{sin(\gamma t|B_k'|)}{|B_k'|}(h'cos\frac{\alpha}{2} + J'cos(\frac{\alpha}{2}+k)))|0\rangle \\
+ (isin\frac{\alpha}{2}cos(\gamma t|B_k'|) + \frac{sin(\gamma t|B_k'|)}{|B_k'|}(h'sin\frac{\alpha}{2} + J'sin(\frac{\alpha}{2}+k)))|1\rangle
\end{aligned}$$

Consider the example where $J = 0$, $h = 1$, $J' = 1$ and $h' = 0$:

$$|\phi(0)\rangle = i|1\rangle$$

Here, $cos\frac{\alpha}{2} = 0$ and $sin\frac{\alpha}{2} = 1$. $|B_k'| = 1$. Which gives:

$$|\phi(t)\rangle = -sin(\gamma t)sink|0\rangle + (cos(\gamma t) + isin(\gamma t)cosk))|1\rangle$$

Considering $\gamma = 1$ and plotting the time evolution of the ground state vector we get Fig.1

Noting the fact that the rate function becomes non-analytic when the time evolved state is perpendicular to the initial state, on the Bloch sphere this can be validated when the time-evolved state is diametrically opposite to the initial state. This occurs when $\vec{B_k}\cdot\vec{B_k'} = 0$. In this case $\vec{B_k} = \{0, 0, 1\}$ and $\vec{B_k'} = \{0, sink, cosk\}$, so $\vec{B_k}\cdot\vec{B_k'} = cosk$ which goes to zero for one value of k when the system size is even.

Consider the time-evolved state $|\phi(t)\rangle$, to make it look like a Bloch vector, factor out the phase factor which lies in the co-efficient of $|0\rangle$ to make the co-efficient of $|0\rangle$ a cosine term. This is accomplished in the following steps (assuming $\gamma = 1$):

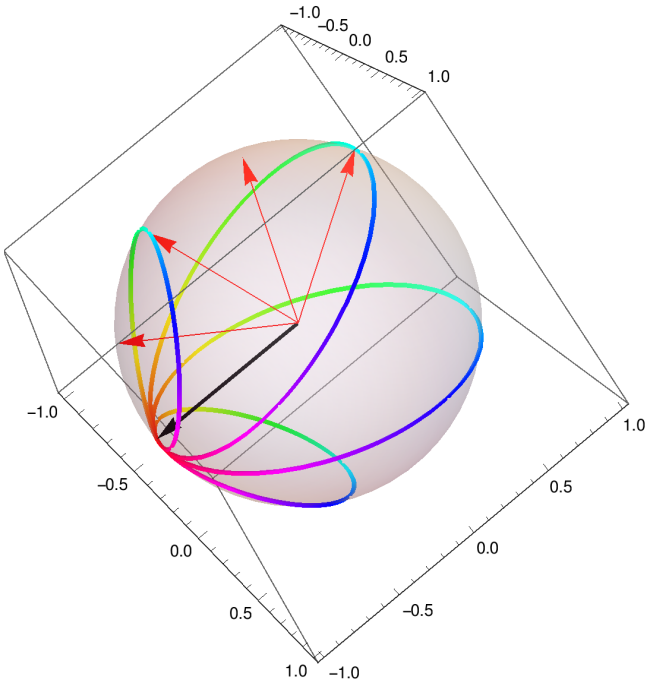


Figure 3: With the system size being 5 and the values of k ranging from $\pi/5$ to $4\pi/5$ in steps of $\pi/5$,the black arrow represents the initial magnetic field vector which is pointing along the z-axis, the other red arrows represent the quenched magnetic field vector and the colored circles shows the time evolution of the ground state on the Bloch sphere. Note that each colored circle starts from where the initial magnetic field is pointing and precesses about the red arrow with time.

$$\vec{B'_k} \cdot \vec{\sigma} = \begin{bmatrix} h' + J'cos(k) & iJ'sin(k) \\ -iJ'sin(k) & -h' - J'cos(k) \end{bmatrix} \tag{58}$$

$$\begin{bmatrix} h' + J'cos(k) & iJ'sin(k) \\ -iJ'sin(k) & -h' - J'cos(k) \end{bmatrix} = U^\dagger \begin{bmatrix} -|B'_k| & 0 \\ 0 & |B'_k| \end{bmatrix} U$$

$$U^\dagger = \begin{bmatrix} cos\phi & isin\phi \\ isin\phi & cos\phi \end{bmatrix}$$

$$\vec{B'_k} \cdot \vec{\sigma}|0\rangle = -|B'_k|(cos2\phi|0\rangle + isin2\phi|1\rangle)$$
$$\vec{B'_k} \cdot \vec{\sigma}|1\rangle = |B'_k|(isin2\phi|0\rangle + cos2\phi|1\rangle)$$

Rewriting $|\phi(t)\rangle$ in this representation, we get:

$$
\begin{aligned}
|\phi(t)\rangle &= (cos\frac{\alpha}{2}cos(t|B'_k|) - isin(t|B'_k|)cos(2\phi - \frac{\alpha}{2}))|0\rangle \\
&+ (isin\frac{\alpha}{2}cos(t|B'_k|) + sin(t|B'_k|)sin(2\phi - \frac{\alpha}{2})))|1\rangle \\
&= A_0 e^{i\phi_0}|0\rangle + A_1 e^{i\phi_1}|1\rangle \\
&= e^{i\phi_0}(A_0|0\rangle + A_1 e^{i(\phi_1-\phi_0)}|1\rangle) \\
A_0^2 &= cos^2\frac{\alpha}{2}cos^2(B'_k t) + sin^2(B'_k t)cos^2(2\phi - \frac{\alpha}{2}) \\
A_1^2 &= sin^2\frac{\alpha}{2}cos^2(B'_k t) + sin^2(B'_k t)sin^2(2\phi - \frac{\alpha}{2}) \\
A_0^2 &+ A_1^2 = 1 \\
A_0 &:= cos\frac{\beta}{2}; \quad A_1 := sin\frac{\beta}{2} \\
tan\phi_0 &= -tan(B'_k t)\frac{cos(2\phi - \frac{\alpha}{2})}{cos\frac{\alpha}{2}} \\
tan\phi_1 &= tan(B'_k t)\frac{sin(2\phi - \frac{\alpha}{2})}{sin\frac{\alpha}{2}}
\end{aligned}
$$

Discarding the overall phase factor, we get:

$$|\phi(t)\rangle = cos\frac{\beta}{2}|0\rangle + e^{i(\phi_1-\phi_0)}sin\frac{\beta}{2}|1\rangle \tag{59}$$

$$\langle\phi(0)|\phi(t)\rangle = cos\frac{\beta}{2}cos\frac{\alpha}{2} + ie^{i(\phi_1-\phi_0)}sin\frac{\beta}{2}sin\frac{\alpha}{2} \tag{60}$$

2.5 Rate function at the critical points:

New conventions : Let $J = 1/2$ and $h = -g_0/2$, $h' = -g_1/2$ and $\phi_k = \theta_k - \theta'_k$, $tan(2\theta_k) = sink/(g0 - cosk)$ and $tan(2\theta'_k) = sink/(g1 - cosk)$ So,

$$
\begin{aligned}
\lambda(t) &= \int_k \frac{dk}{2\pi} log(|(cos^2\phi_k + sin^2\phi_k e^{-2itB'_k})|^2) \\
z &= cos^2\phi_k + sin^2\phi_k e^{-2itB'_k} \\
z\bar{z} &= |(cos^2\phi_k + sin^2\phi_k e^{-2itB'_k})|^2 \\
\frac{d\lambda(t)}{dt} &= \int_k \frac{dk}{2\pi}\frac{z\partial_t\bar{z} + \bar{z}\partial_t z}{z\bar{z}} \\
&= Re[\int_k \frac{dk}{\pi}\frac{\partial_t\bar{z}}{\bar{z}}] \\
&= Re[\int_k \frac{dk}{\pi}\frac{\partial_t(cos^2\phi_k + sin^2\phi_k e^{-2itB'_k})}{cos^2\phi_k + sin^2\phi_k e^{-2itB'_k}}] \\
&= Re\int_k \frac{dk}{\pi}\frac{-2iB'_k sin^2\phi_k e^{-2itB'_k}}{cos^2\phi_k + sin^2\phi_k e^{-2itB'_k}}
\end{aligned}
$$

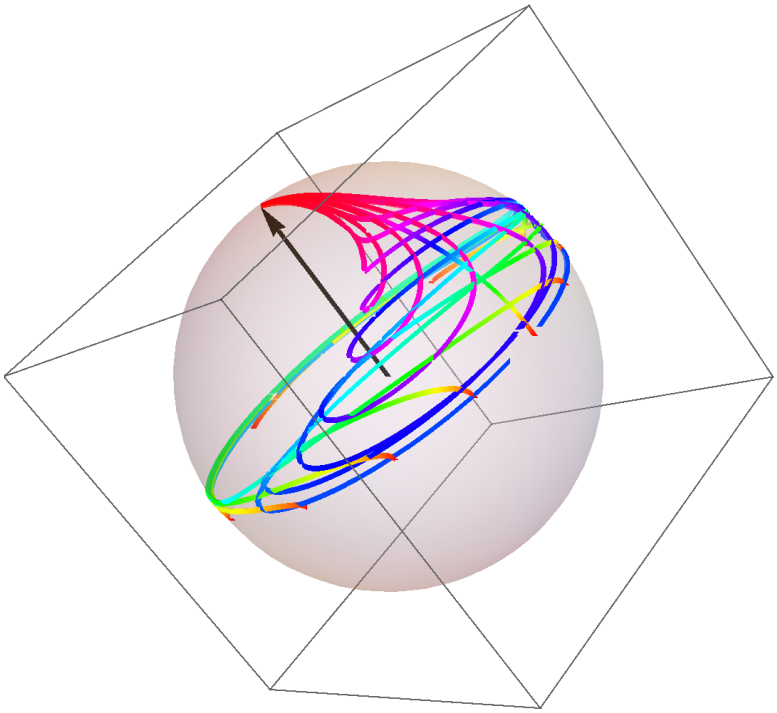


Figure 4: g0 = 0.9,g1 = 2.0

Simplifying further and evaluating the derivative at the critical points $t = t^*$ and $k = k^*$ where $t^* = \dfrac{\pi}{2B'_{k^*}}$ and $cosk^* = \dfrac{1+g_0g_1}{g_0+g_1}$.Also note that $\phi_{k^*} = \pi/4$ At the critical points, the integral becomes divergent. To resolve this issue we use the Sokhotski-Plemelj theorem which states :

$$\lim_{\epsilon\to 0^+}\int_a^b \frac{f(x)}{x\pm i\epsilon}dx = \mp i\pi f(0) + \mathrm{P}\int_a^b \frac{f(x)}{x} \tag{61}$$

We evaluate the derivative of the rate function at $t = t^* + \epsilon$ around the critical value of k i.e., k^* to know the behaviour of the slope of the rate function around the critical point.

$$\frac{d\lambda(t)}{dt}\,|_{t=t^*+\epsilon} = Re\int_k \frac{dk}{\pi}\frac{-2iB'_k sin^2\phi_k e^{-2itB'_k}}{cos^2\phi_k + sin^2\phi_k e^{-2itB'_k}}$$

The denominator in the integral goes to zero at the critical points. So taking the first order approximation in k about k^* while safely evaluating the numerator at k^* we get:

$$\frac{d\lambda(t)}{dt}\,|_{t=t^*+\epsilon} = Re\int_k \frac{dk}{\pi}\frac{-2iB'_{k^*}(sin^2\pi/4)e^{-2iB'_{k^*}\pi/(2B'_{k^*})}}{cos^2\phi_k + sin^2\phi_k e^{-2itB'_k}}$$

Taylor expansion of the terms in the denominator:

$$
\begin{aligned}
cos^2\phi_k &= cos^2\phi_{k^*} - 2sin\phi_{k^*}cos\phi_{k^*}\partial_k\phi_k\delta k = \frac{1}{2} - \partial_{k^*}\phi_k\delta k \\
sin^2\phi_k &= sin^2\phi_{k^*} + 2sin\phi_{k^*}cos\phi_{k^*}\partial_k\phi_k\delta k = \frac{1}{2} + \partial_{k^*}\phi_k\delta k \\
e^{-2i(t^*+\epsilon)B'_k} &= e^{-i\frac{\pi B'_k}{B'_{k^*}}-2i\epsilon B'_k} = e^{-i\frac{\pi B'_{k^*}+\pi\partial_k B'_k\delta k}{B'_{k^*}}} e^{-2i\epsilon(B'_{k^*}+\partial_k B'_k\delta k)} \\
&= -e^{-i\pi\frac{\partial_k B'_k\delta k}{B'_{k^*}}} e^{-2i\epsilon B'_{k^*}} = -(1 - i\pi\frac{\partial_k B'_k\delta k}{B'_{k^*}} - 2i\epsilon B'_{k^*}) \\
cos^2\phi_k + sin^2\phi_k e^{-2itB'_k} &= \frac{1}{2} - \partial_{k^*}\phi_k\delta k - (\frac{1}{2} + \partial_{k^*}\phi_k\delta k)(1 - i\pi\frac{\partial_k B'_k\delta k}{B'_{k^*}} - 2i\epsilon B'_{k^*}) \\
&= (-2\partial_{k^*}\phi_k + i\pi\frac{\partial_k B'_k}{2B'_{k^*}})\delta k + i\epsilon B'_{k^*} \\
(-2\partial_{k^*}\phi_k + i\pi\frac{\partial_k B'_k}{2B'_{k^*}}) &= c_R + ic_I
\end{aligned}
$$

$$
Re\int_k \frac{dk}{\pi}\frac{iB'_{k^*}}{cos^2\phi_k + sin^2\phi_k e^{-2itB'_k}} = Re\int_k \frac{dk}{\pi}\frac{\frac{iB'_{k^*}}{c_R+ic_I}}{\delta k + \frac{i\epsilon B'_{k^*}}{c_R+ic_I}}
$$

$$
\begin{aligned}
\frac{B'_{k^*}}{c_R+ic_I} &= d_R + id_I \\
&\Rightarrow Re\int_k \frac{d\delta k}{\pi}\frac{(d_R+id_I)}{\delta k + \epsilon(d_R+id_I)} \\
&= Re\int_k \frac{d\delta k}{\pi}\frac{\frac{i(d_R+id_I)}{d_R}}{\frac{(\delta k-\epsilon d_I)}{d_R} + i\epsilon} \\
\delta k' &= \frac{(\delta k - \epsilon d_R)}{d_I} \\
&\Rightarrow \lim_{\epsilon\to 0} Re\int_{\delta k'} \frac{d\delta k'}{\pi}\frac{i(d_R+id_I)}{\delta k' + i\epsilon} \\
&= sgn(\epsilon)Re(-d_R - id_I) = -sgn(\epsilon)d_R
\end{aligned}
$$

$$d_R = \frac{B'_{k^*} c_R}{c_R^2 + c_I^2} = \frac{B'_{k^*}(-2\partial_{k^*}\phi_k)}{(-2\partial_{k^*}\phi_k)^2 + (\pi \frac{\partial_k B'_k}{2B'_{k^*}})^2}$$

$$\theta_k = \frac{1}{2} tan^{-1}(sink/(g0 - cosk))$$

$$\partial_k \theta_k = \frac{1}{2}\frac{(g0 - cosk^*)^2}{B_{k^*}^2} = \frac{1}{2}\frac{(g0 - (\frac{1 + g0g1}{g0 + g1}))^2}{B_{k^*}^2} = \frac{1}{2}\frac{(\frac{g0^2 - 1}{g0 + g1})^2}{B_{k^*}^2}$$

$$\partial_k \theta'_k = \frac{1}{2}\frac{(\frac{g1^2 - 1}{g0 + g1})^2}{B'^2_{k^*}}$$

$$\partial_k B_k = \frac{g0 sink^*}{B_{k^*}}$$

$$\partial_{k^*}\phi_k = \partial_{k^*}(\theta_k - \theta'_k) = \frac{1}{2}\frac{(\frac{g0^2 - 1}{g0 + g1})^2}{B_{k^*}^2} - \frac{1}{2}\frac{(\frac{g1^2 - 1}{g0 + g1})^2}{B'^2_{k^*}}$$

Near the critlcal $k = k^*$, taking the differnce in the slopes across the critical time $t = t^*$ we get:

$$\Delta \frac{d\lambda(t)}{dt} \mid_{t=t^*} = -2d_R$$

$$c_R = \frac{\mathrm{g0} + \mathrm{g1}}{\mathrm{g0} - \mathrm{g1}}$$

$$c_I = \frac{\pi \mathrm{g1} n (\mathrm{g0} + \mathrm{g1}) \sqrt{1 - \frac{(\mathrm{g0g1}+1)^2}{(\mathrm{g0}+\mathrm{g1})^2}}}{2\left(\mathrm{g1}^2 - 1\right)(\mathrm{g1} - \mathrm{g0})}$$

$$\therefore \Delta \frac{d\lambda(t)}{dt} \mid_{t=t^*} = \frac{4(\mathrm{g0} - \mathrm{g1}) \sqrt{-\frac{\left(\mathrm{g1}^2 - 1\right)(\mathrm{g0} - \mathrm{g1})}{\mathrm{g0} + \mathrm{g1}}}}{(\mathrm{g0} + \mathrm{g1})\left(4 - \frac{\pi^2 \left(\mathrm{g0}^2 - 1\right) \mathrm{g1}^2 n^2}{\left(\mathrm{g1}^2 - 1\right)(\mathrm{g0} + \mathrm{g1})^2}\right)}$$

From Figure 4, we can infer that the rate function reaches a saturation value at larger times. Whereas, the finite size systems don't keep oscillating and don't seem to saturate.

2.6 Numerical simulations

We perform numerical simulations for the spin representation of the transverse field ising model and compare the obtained results with the analytic solutions.

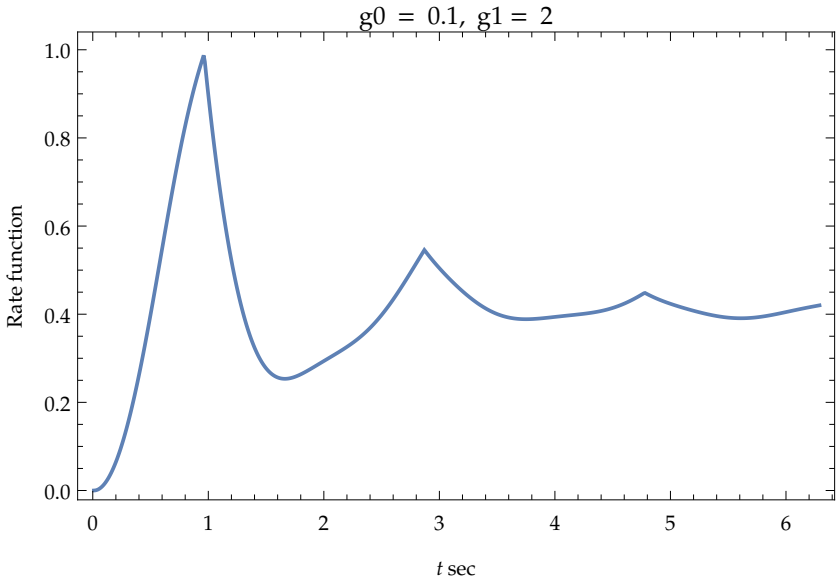


Figure 5: Numerically integrating the logarithm of the Loschmidt amplitude in order to obtain the rate function

3 Clock model $\mathscr{Z}-3$ invariant spin chains

3.1 Defining the Hamiltonian

Consider three possible states for a $\mathscr{Z}-3$ invariant clock model $|\uparrow\rangle, |\searrow\rangle$ and $|\swarrow\rangle$

Consider the operators

$$\sigma = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{bmatrix}, \tau = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \tag{62}$$

Here σ measures the state of the spin while τ rotates it by $\frac{2\pi}{3}$. Now not worrying about the physical significance for the moment we try to construct a hamiltonian by drawing analogy with the transverse field Ising model.

The hamiltonian for the transverse field Ising model is :

$$H_{IM} = -J\sum_{n=1}^{N-1}\sigma_n^z\sigma_{n+1}^z - f\sum_{n=1}^{N}\sigma_n^x \tag{63}$$

$$H_3 = -J\sum_{n=1}^{N-1}(\sigma_n\sigma_{n+1}^\dagger e^{i\hat{\phi}} + \sigma_n^\dagger\sigma_{n+1}e^{-i\hat{\phi}}) - f\sum_{n=1}^{N}(\tau_n e^{i\phi} + \tau_n^\dagger e^{-i\phi}) \tag{64}$$

References

[1] Quantum Field Theory and Condensed Matter: An Introduction (Cambridge Monographs on Mathematical Physics) 1st Edition, R.Shankar

[2] Dynamical quantum phase transitions: a review,Markus Heyl

[3] M.Heyl,PRL 110, 135704 (2013)

[4] Parafermionic edge zero modes in Z n -invariant spin chains, Paul Fendley

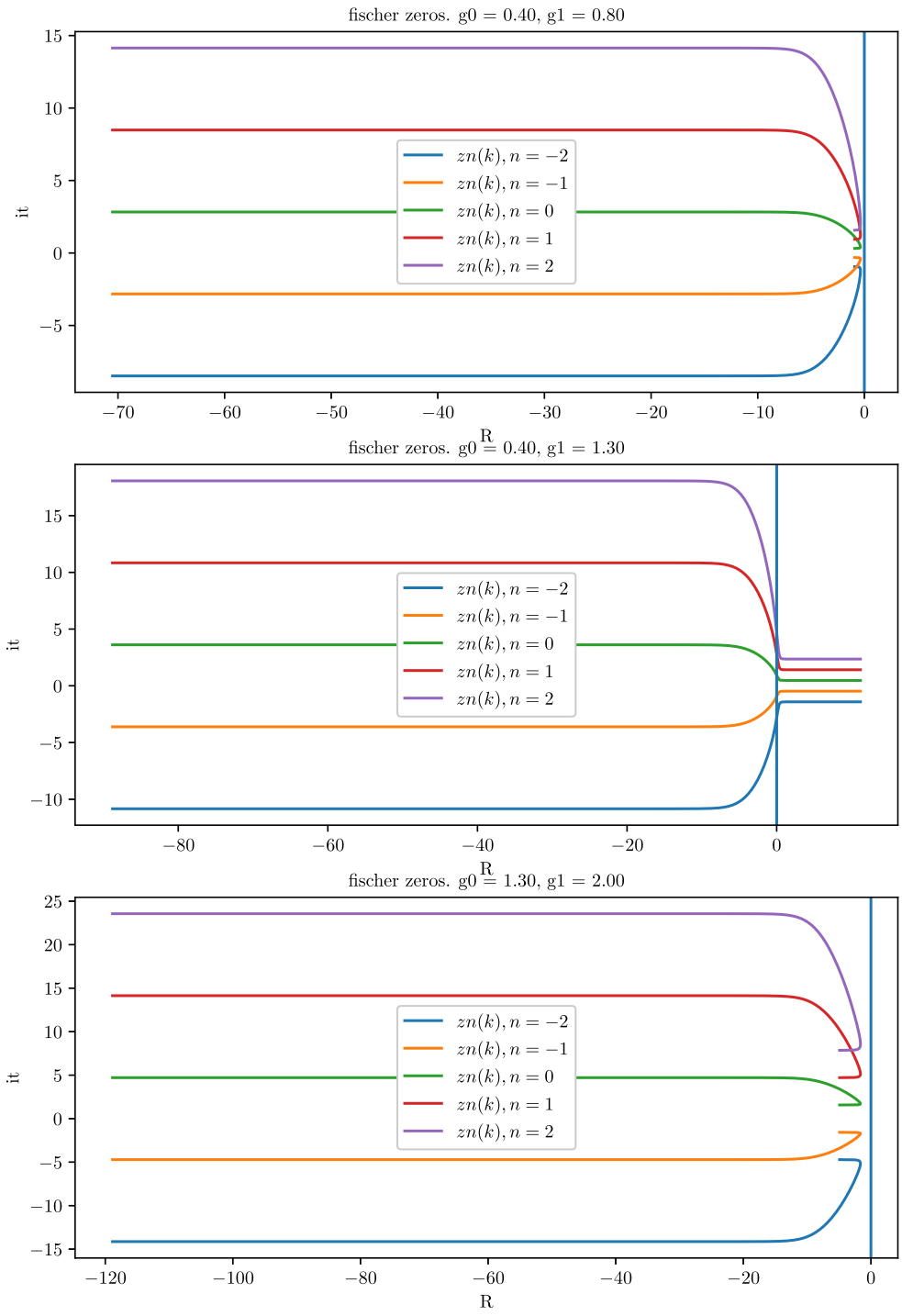


Figure 6: Plotting the zeroes of the Loschmidt amplitude in the complex plane. These plots clearly shows that dynamical phase transitions occurs only when quenches are across the critical point (g = 1)

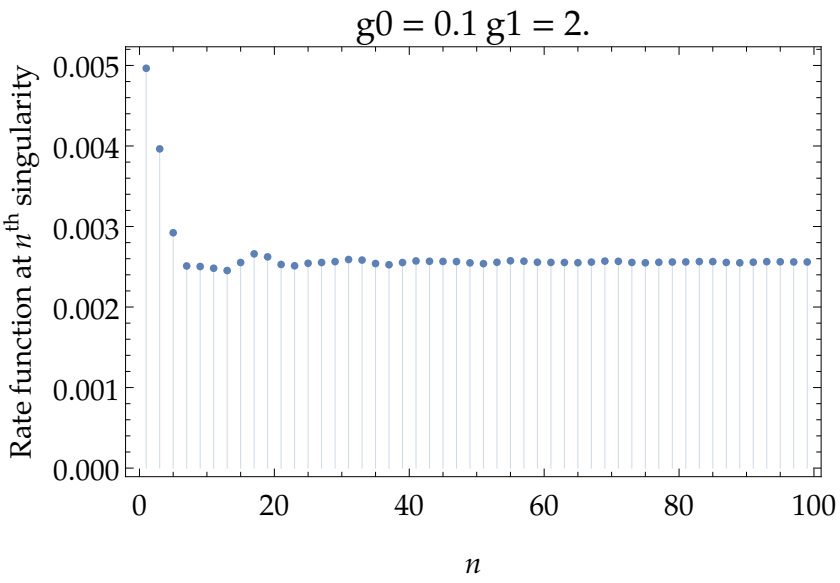


Figure 7: .

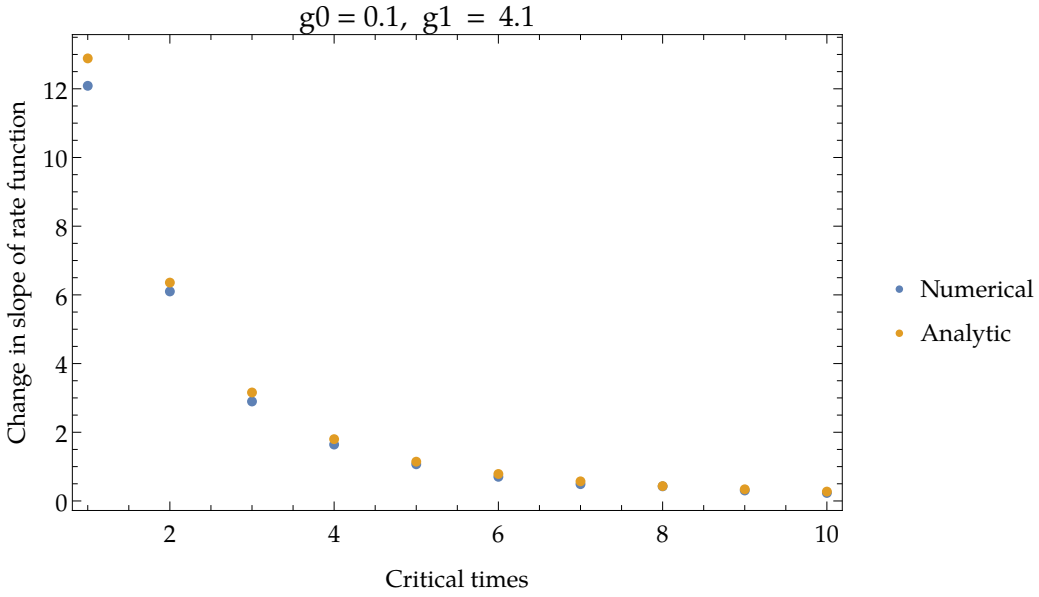


Figure 8: .

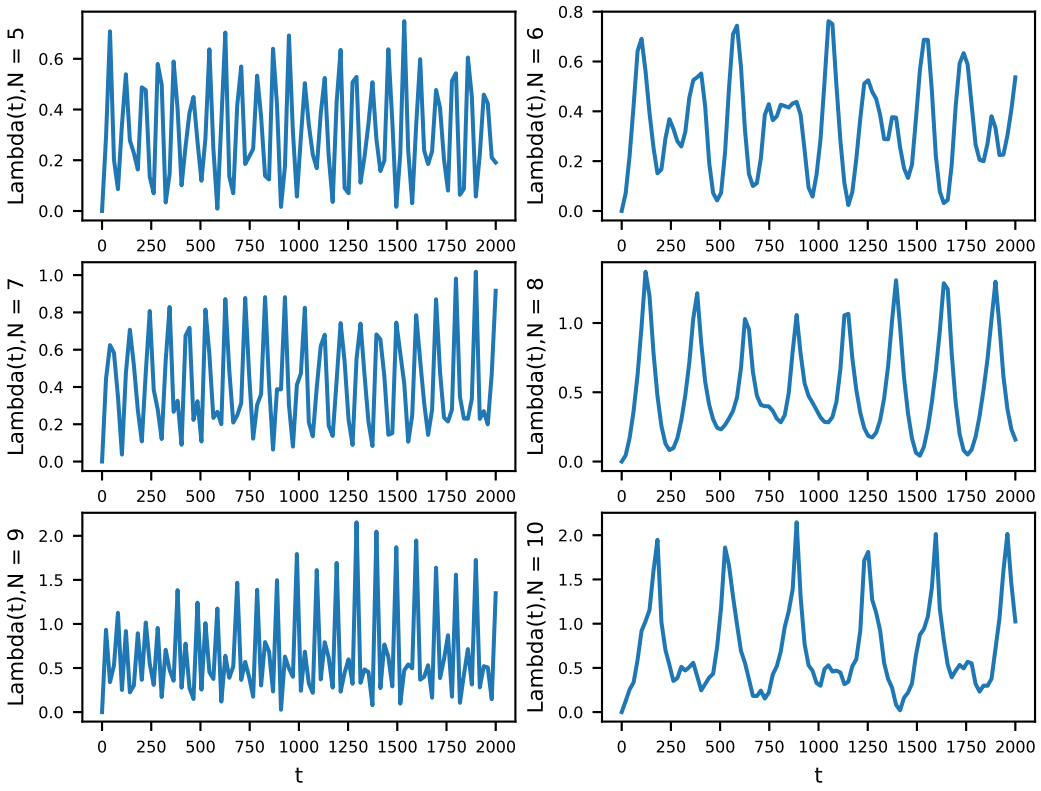


Figure 9: Numerical calculation of rate function for g0 = 0.1, g1 = 2.1. The value of the rate function clearly does not saturate with time.

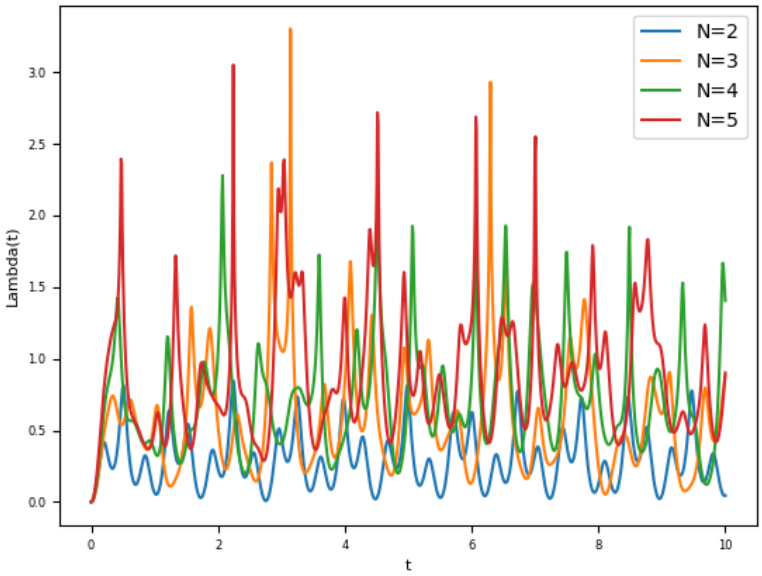


Figure 10: Numerical simulation of the Potts model